

E. coli* at Toronto's Beaches: Are Some Beaches Less Safe Than Others?

Marie Curtis Park East and Sunnyside exceed the swimming limit on a third of summer days, more than seven times as often as Gibraltar Point

Sean Murphy

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Sentence 1: the general area (Toronto's beaches and the daily *E. coli* warning). Sentence 2: data and method (98,370 samples at ten beaches, 2007 to 2026, from Open Data Toronto; days over the limit, a same-day model, yesterday against today). Sentence 3: headline result (Marie Curtis Park East and Sunnyside over the limit on about a third of days, more than seven times Gibraltar Point; yesterday's samples catch 37% of days over the limit). Sentence 4: implication (knowing which beach you are at tells a swimmer more than yesterday's reading).

1 Introduction

[Why *E. coli*] *E. coli* is a sign of faecal pollution. It lives in the gut of people and other warm-blooded animals, so finding it in lake water means that sewage, storm-water runoff or bird droppings have arrived recently, bringing the viruses and bacteria that make swimmers ill [1]. Swimming in water above Toronto's standard raises the risk of ear, eye, nose, throat and skin infections, and swallowed water can cause stomach illness. Young children, older people and people with weakened immune systems are most at risk [2].

[Questions and gap] Placeholder: the two questions (do some beaches go over the limit more often; how well does yesterday's result predict today's) and the gap: no published comparison of the beaches over 20 years, and no check of the one-day delay.

[Findings] Placeholder: 34% and 32% of days at Marie Curtis Park East and Sunnyside against 4% at Gibraltar Point; a warning based on yesterday catches 37% of the days over the limit.

*Code and data are available at: <https://github.com/seandata8/toronto-beaches-ecoli>.

[Why it matters] Placeholder: which beach matters more than which day, and the posting is a weak guide.

[Roadmap] Placeholder: Section 2 describes the data, how a sample becomes a number, and the methods; Section 3 gives the results; Section 4 what they mean and their limits. The appendices cover cleaning and the check on simulated data.

2 Data

2.1 Source and context

[Data source] To reduce water-borne illness among swimmers, Toronto Public Health, Toronto Water, the police Marine Unit and the City’s parks division jointly sample the water at the city’s ten supervised beaches every day from Victoria Day to Labour Day [3]. Samples are taken at four to six fixed sites along each beach (Figure 1), and when the laboratory count of *E. coli* at a beach is too high, Toronto Public Health posts signs warning against swimming [4]. The results are published on the City’s open data portal [3,5] and form the data for this paper.

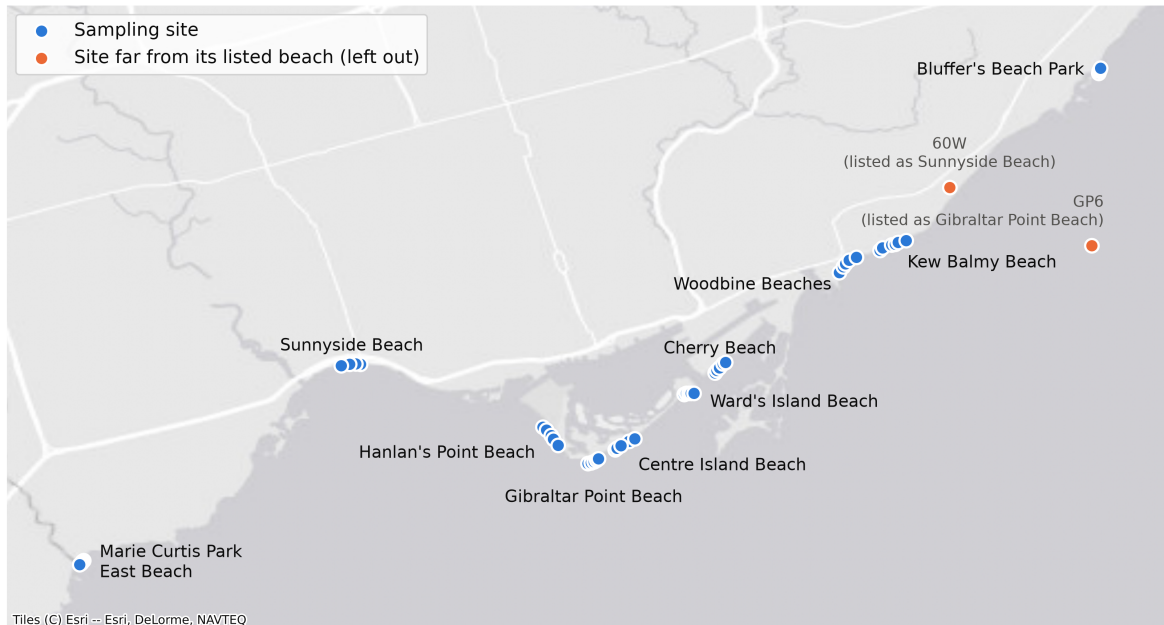


Figure 1. Each of Toronto’s ten swimming beaches is sampled at four to six sites along the shore. Blue points are the 50 sites this paper uses. The two orange points, sites 60W and GP6, were added in 2026 and sit 14 to 16 km from the beaches they are listed under, so they are left out. Map tiles from Esri.

[Swimming limit] Toronto’s standard of 100 *E. coli* per 100 mL is half that of Ontario and Health Canada: Health Canada raised its guideline from 100 to 200 in 2012, and Ontario follows it, whereas Toronto retained the earlier, stricter value [1,2]. Because 100 is the level at which Toronto posts warnings, it is the limit used throughout this paper.

[Ethics and alternatives] Because the data describe water quality rather than people, they raise no privacy concerns; the ethical stakes lie instead in the advice built on them, since a needless warning costs swimmers a beach day and a missed one exposes them to illness [6]. The City also offers each beach’s daily swimming condition for download [4], but the open data were used here because they give every individual result, allowing each day’s status to be calculated the same way for all twenty years.

[Software] The analysis uses Python [7] with Polars [8], NumPy [9], statsmodels [10], pointblank [11], matplotlib [12] and contextily [13], and the paper is written in Quarto [14].

2.2 Measurement

[Columns] Each row of the dataset records one site on one day: the beach, a site code and its location (Figure 1), the date of collection and the *E. coli* result. To obtain each result, staff collect 200 mL of water about 30 cm below the surface at each of the beach’s sites (five at most beaches, six at Kew Balmy and four at Sunnyside), and a Public Health Ontario laboratory counts the *E. coli* by membrane filtration, in which bacteria trapped on a filter grow overnight into visible colonies that are counted and scaled to a volume of 100 mL [15]. Because the process takes about a day, a posted result describes the previous day’s water [2,4]. When no sample could be taken, usually because of bad weather, the result was left blank [2]; 3.5% of rows are blank.

[Floor and ceiling] The laboratory’s reporting limits compress the data at both ends (Figure 2). Results below 10 are rarely reported, so the 49% of results at exactly 10 are best read as “10 or fewer”. From 2018, high results were increasingly reported as exactly 1,000; earlier results reached 15,820, whereas none exceeded 1,000 in 2026, which suggests that a reporting ceiling was introduced. Because both limits narrow the range, differences among the cleanest beaches and the severity of the worst days are likely understated. The ceiling can also hide a day over the limit: since 2018, 34 days at a beach included a result of 1,000 yet had a daily geometric mean (Section 2.3) below 100, and on about 8 of them a true result of 2,000 or less would have put the day over. The true number of days over the limit since 2018 may therefore be slightly higher than the number calculated from these data.

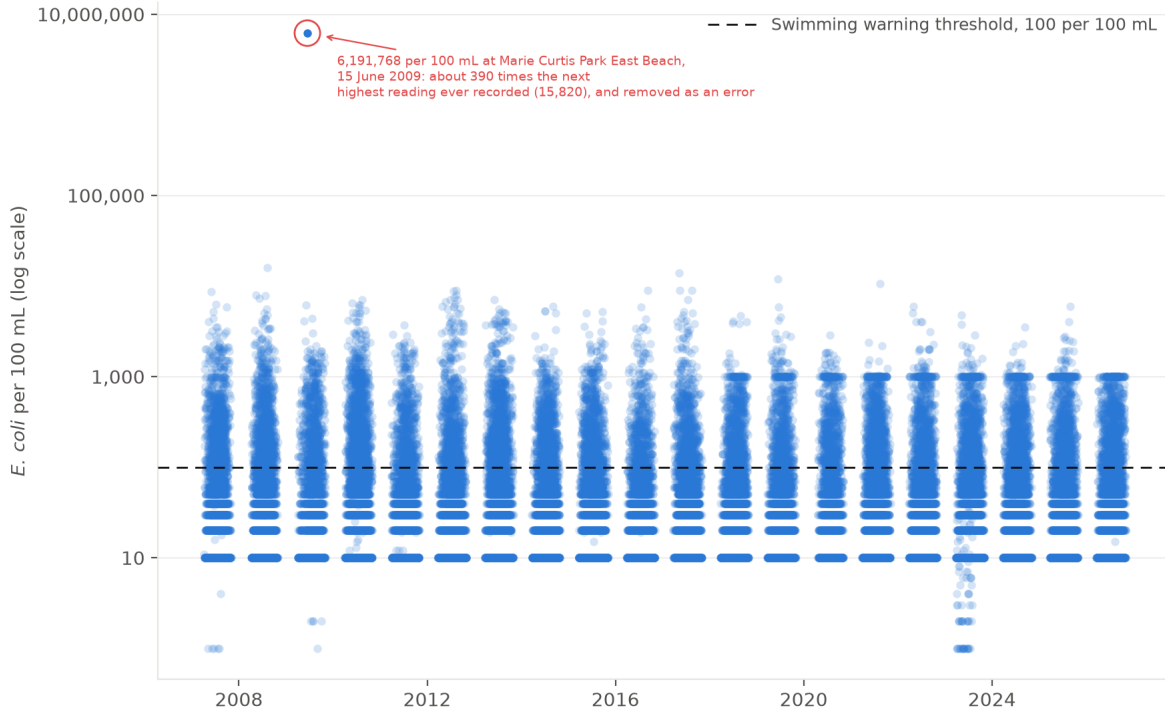


Figure 2. Every result the City has published, 2007 to 2026. One point is one water sample (98,991 in total), before any cleaning. The vertical axis is logarithmic, so each gridline is 100 times the one below. Points are spread sideways within each year so that they can be seen; the *E. coli* count itself is never moved. Most results are reported in multiples of 10, which is why the low readings fall into separate rows. The laboratory reports no results as zero, and “no colonies detected” is assumed to be reported as 10. From 2018 on, many results are reported as exactly 1,000, and these are assumed to be readings of 1,000 or more. The dashed line marks 100 *E. coli* per 100 mL, the level at which the City posts a swimming warning. The ringed point is the only reading the analysis removes.

[Cleaning] Before analysis, three groups of results were removed: one implausible reading (ringed in Figure 2), the results from two sites that lie far from the beaches they are listed under (Figure 1), and results from dates outside the swimming season. The remaining 98,370 results come from 50 sites on 2,019 days across 20 years; Appendix A gives the details.

[Summary statistics] Most results are low, but two beaches stand apart (Table 1). At eight of the ten beaches the median result (the middle value when all results are ranked) is 10 or 20, and at Gibraltar Point, the cleanest, 61% of results are at the floor of 10 and only 6% exceed 100. At Marie Curtis Park East and Sunnyside, by contrast, the medians are 50 and 60, only 27% and 20% of results are at the floor, and 35% and 34% exceed 100, about six times the share at Gibraltar Point.

Table 1. Results at each beach, 2007 to 2026. A result is the number of *E. coli* per 100 mL in one water sample. Ten is usually the lowest value the laboratory reports, and 100 is Toronto’s limit for swimming. The 90th percentile is the level that one result in ten exceeds. Beach-days are days with at least four results. Beaches run from the largest share of results above 100 to the smallest.

Beach	Sites	Results	Median	90th percentile	At 10	Above 100	Beach-days
Marie Curtis Park East	5	9,612	50	590	27%	35%	1,925
Sunnyside	4	8,006	60	400	20%	34%	1,986
Centre Island	5	9,645	20	220	40%	19%	1,929
Kew Balmy	6	12,047	20	250	47%	19%	2,011
Bluffer’s Park	5	10,006	10	150	55%	13%	2,001
Woodbine	5	10,035	10	110	59%	11%	2,005
Ward’s Island	5	9,757	10	110	58%	10%	1,952
Cherry	5	10,012	10	90	58%	9%	2,003
Hanlan’s Point	5	9,689	10	80	61%	8%	1,939
Gibraltar Point	5	9,561	10	60	61%	6%	1,911

2.3 Methods

[Beach-days] Because Toronto’s limit applies to the geometric mean of a beach’s results on a day rather than to any single result [16], the analysis works with beach-days. The geometric mean of n results is the n th root of their product, or equivalently the average of their logarithms converted back. It suits counts that span orders of magnitude, because a single extreme result does not dominate it: results of 10, 10, 10, 10 and 1,000 give a geometric mean of 25, against an ordinary average of 208. A beach-day was counted only when it had at least four results, rather than five, so that Sunnyside (four sites) is included. This gives 19,662 beach-days, of which 2,816 (14%) were over the limit. A day over the limit is not necessarily a day on which a warning was posted—the data do not record postings. Saleem et al. describe posting decisions based on a geometric mean over two days [6]. In addition, a posted warning reflects the previous day’s water quality, because testing takes about 24 hours (Section 2.2).

[Model] To test whether the differences reflect the beaches rather than the weather on the days each was sampled, each beach was compared only with the other beaches sampled on the same day. For beach b on day d , the model is

$$\log_{10} G_{bd} - \bar{L}_d = \beta_b + \varepsilon_{bd}, \quad (1)$$

where, in Equation 1, G_{bd} is the beach’s geometric mean that day, $\bar{L}_d$ is the average of $\log_{10} G$ over all beaches sampled that day, β_b is the beach’s average gap from the others, and ε_{bd} is

the day-to-day variation the model does not explain. Subtracting each day’s average removes anything that affects all beaches at once, such as a city-wide storm. Because the model works on a log scale, a gap of $\beta_b = 0.3$ means about twice the *E. coli* of the average beach on the same day ($10^{0.3} \approx 2$).

[Model] The standard error of each β_b was calculated from the variation between calendar months rather than between days, because a wet week raises a beach’s levels for several days running; this roughly doubles the width of the intervals compared with treating every day as independent. Each estimate is reported with a 95% interval: if the analysis were repeated over many different runs of summers, about 95% of such intervals would contain the true gap. Two beaches were judged to differ when a gap as large as the one observed would arise less than 5% of the time if they were in fact alike. Testing all 45 pairs this way would produce about two false differences by chance, so the comparisons were corrected by Holm’s method, which raises the bar each comparison must clear so that the chance of any false difference stays near 5%.

[Consecutive days] Because results take about a day, the advice on display at a beach on any day, called here the posting day, is based on the samples taken the day before. To measure how often that advice was right, each posting day was paired with the previous day at the same beach, using only days that were consecutive (18,998 pairs). A warning was taken to be posted if the previous day’s geometric mean was over the limit, and no warning if it was not. That advice was then compared with the water sampled on the posting day itself to assess if the advice matched the water quality on the posting day.

3 Results

[Beach comparison] Placeholder: Figure 3; shares from 34.1% and 31.8% down to 4.4%; the same top two in 19 of 20 years (exception 2021); splitting at 2018 barely moves the shares.

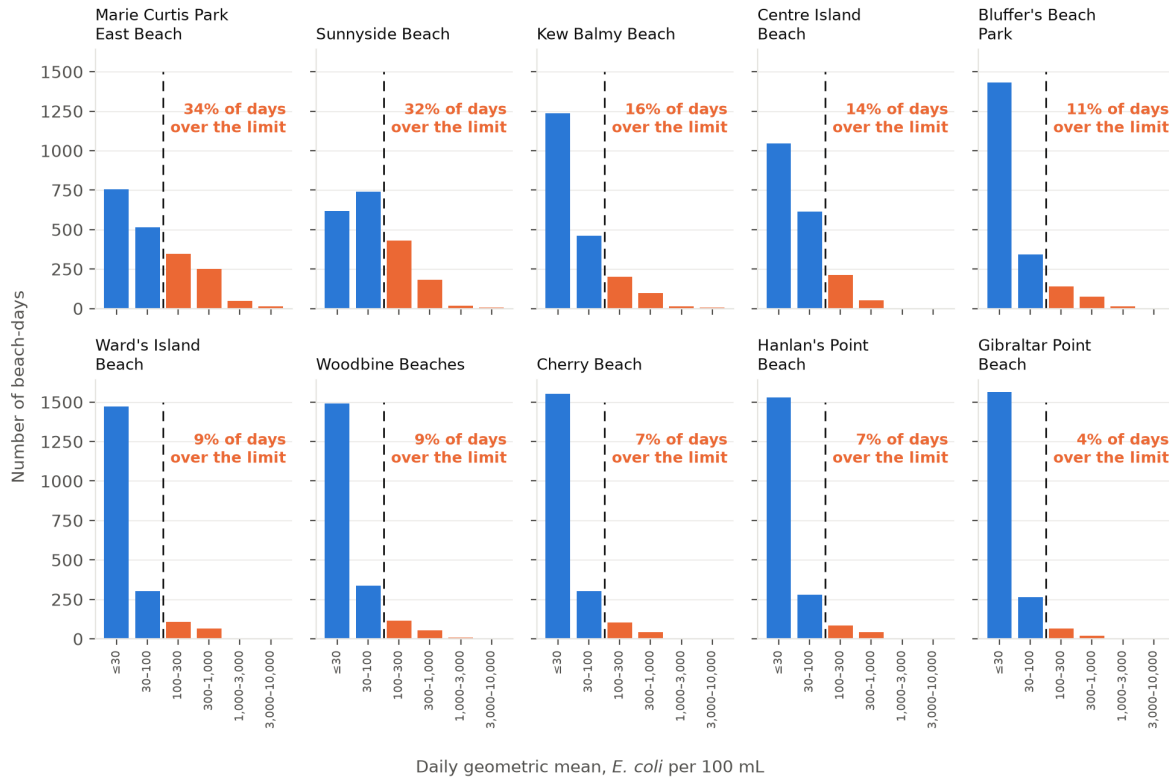


Figure 3. How often each beach went over the limit, 2007 to 2026. Each bar counts the beach-days whose geometric mean falls in that range (19,662 days in total; days with fewer than four results are left out). Orange bars, right of the dashed line, are days above Toronto's limit of 100 *E. coli* per 100 mL. The lowest bin includes the 17 days whose mean falls below the reporting floor of 10. Panels run from the largest share of days over the limit to the smallest.

[Model estimates] Placeholder: Figure 4; Marie Curtis and Sunnyside 2.2 times the average beach on the same day, not different from each other, different from every other beach; 31 or 32 of 45 pairs differ; the reporting limits make the gaps look smaller than they are. Centre Island ranks above Kew Balmy in the model but below it in share of days, and why.

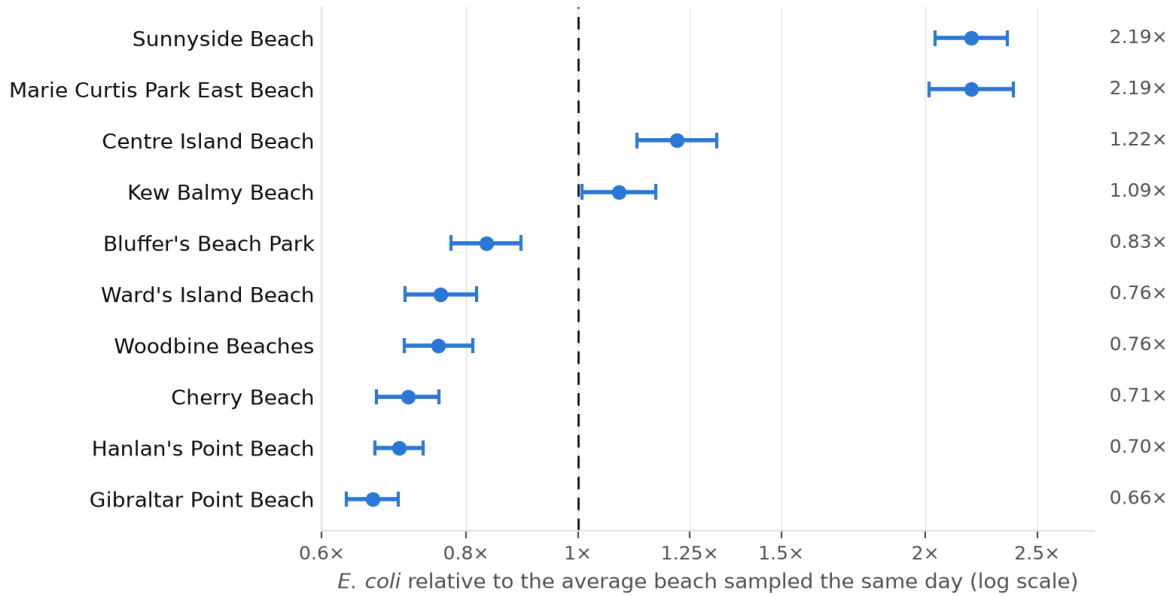


Figure 4. Placeholder caption: each beach’s typical level against the average beach on the same day, on the log10 scale, with 95% intervals. The gaps are understated: the reporting floor and ceiling shrink them to about two thirds of their true size.

[Yesterday vs today] Placeholder: Table 2; name the four outcomes the table shows (correct warning, false warning, missed warning, correct no-warning), each tied to the posting day’s water; avoid the term “confusion matrix” or explain it; of the posting days with water over the limit, 37% had a warning posted; of the warnings posted, 37% were correct; 14% of posting days have water over the limit overall, 37% after a warning and 10% with no warning (about 3.5 times as likely). Leave out overall accuracy.

Table 2. Advice posted at each beach, based on the previous day’s samples, against the water on the posting day, 2007 to 2026 (18,998 pairs of consecutive days). A warning is posted when the previous day’s geometric mean was above Toronto’s limit of 100 *E. coli* per 100 mL. Each cell gives the number of pairs and its share of all pairs.

Advice	Water over the limit on the posting day	Water under the limit
Warning posted	1,002 (5.3%)	1,712 (9.0%)
No warning posted	1,700 (8.9%)	14,584 (76.8%)

4 Discussion

[Summary] Placeholder: brief summary of the data and results.

[Why beaches differ] Placeholder: Etobicoke Creek at Marie Curtis, the Humber River near Sunnyside, storm-water outfalls, island against mainland beaches; analyses of the data that could test these. Hypotheses, not tested here.

[Improving warnings] Placeholder: the current testing misses most bad days and posts unnecessary warnings most of the time; a prediction model based on weather; qPCR for same-day testing.

[Weaknesses] Placeholder: days over the limit are computed, not recorded postings; the floor and ceiling understate the gaps; no weather data; Sunnyside's four sites add noise (effect under one percentage point in simulation).

[Next steps] Placeholder: compare days over the limit with the warnings actually posted: the City's "Beach Condition Data for Researchers" download (JSON) may record them.

Appendix

A Data cleaning

[Cleaning details] Placeholder: downloaded 22 September 2026; 102,534 rows from 3 June 2007 to 16 September 2026, 98,991 with a result; every removal with counts and reasons (1 implausible reading, 220 results from sites 60W and GP6, 400 results outside the season from Victoria Day to Labour Day) (the reading of 6,191,768; sites 60W and GP6 with a footnote on the City's reply and the download date of 22 September 2026; dates outside the season; days with fewer than four results). Tests run on the cleaned data.

B Checking the method on simulated data

[Simulation check] Placeholder: simulated data with known beach differences, the floor and ceiling; the analysis recovers the ordering; censoring leaves about 65% of a true gap; the site-count effect.

C Pairwise comparisons

[Pairwise table] Placeholder: the 45 pairs with Holm-corrected p-values, and the check with partial months merged. Table from `other/results/beach-pairwise-differences.csv` and `beach-pairwise-grouping-check.csv`.

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